

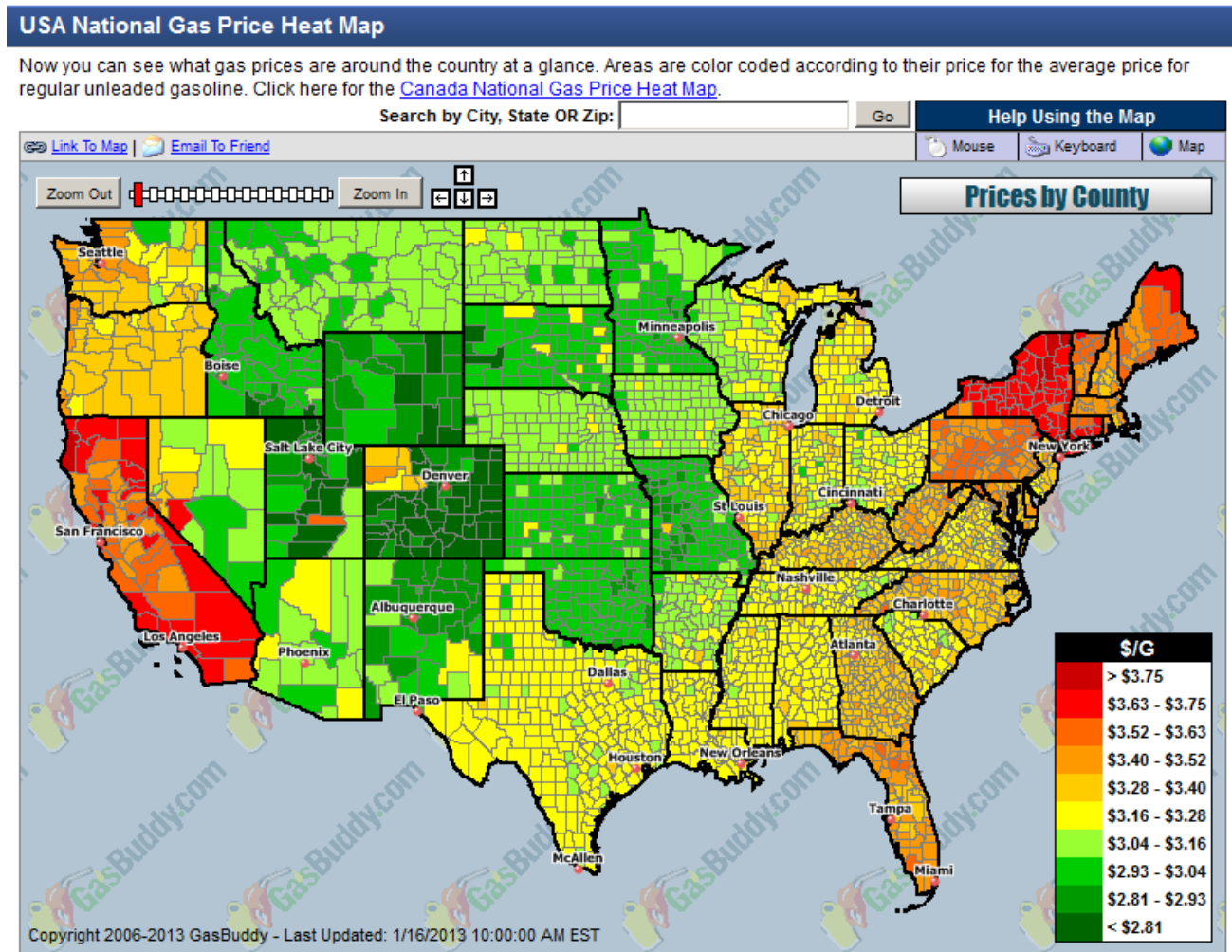
# “Vertical Relationships and Competition in Retail Gasoline Markets”

Justine Hastings, AER, 2004

- Does the market structure affect retail gasoline prices in a market?
- Does vertical integration in retail gasoline markets affect prices/competition?

# Policy Motivation

- Substantial gas price variation
- Policymakers sometimes believe correlated with vertical integration
- But could be driven by integrated refiners buying independents



# Research Questions in More Detail

- Do certain vertical relationships make the market more competitive?
- Types of Vertical Relationships b/t Refiners and Retail Stations
  - 1) Company operated (co-op) by refiner
    - Refiner sets price
  - 2) Lessee-dealer
    - Lessee sets price but contracted to buy from refiner at a specific wholesale price
  - 3) Dealer-owned
    - Dealer sets price & buys branded gas from “jobber”
  - 4) Independent
    - Does not necessarily sell branded gas

# Research Design

- Possible Ideal experiments?
  - Randomly convert some stations from independents to vertically integrated, branded stations
    - Demand-side interpretation?
  - Randomly assign some stations to compete with independents & others to compete with co-op or dealer owned
- “Quasi-experiment”
  - 260 independent Thrifty stations bought by ARCO in Sept. 1997 & converted to leese-dealer, company owned, & dealer owned
    - Sign changed from “Thrifty” to “ARCO”
    - No other changes made to stations (e.g. remodeling)
  - If conversion “randomly assigned” (or at least exog to unobserved determinants of prices), can interpret effect on prices as causal

# Other (weaker) strategies

- Why not use cross-section of markets, some including independents and some not?
- Why not use time series of a market that observes entry of independents?

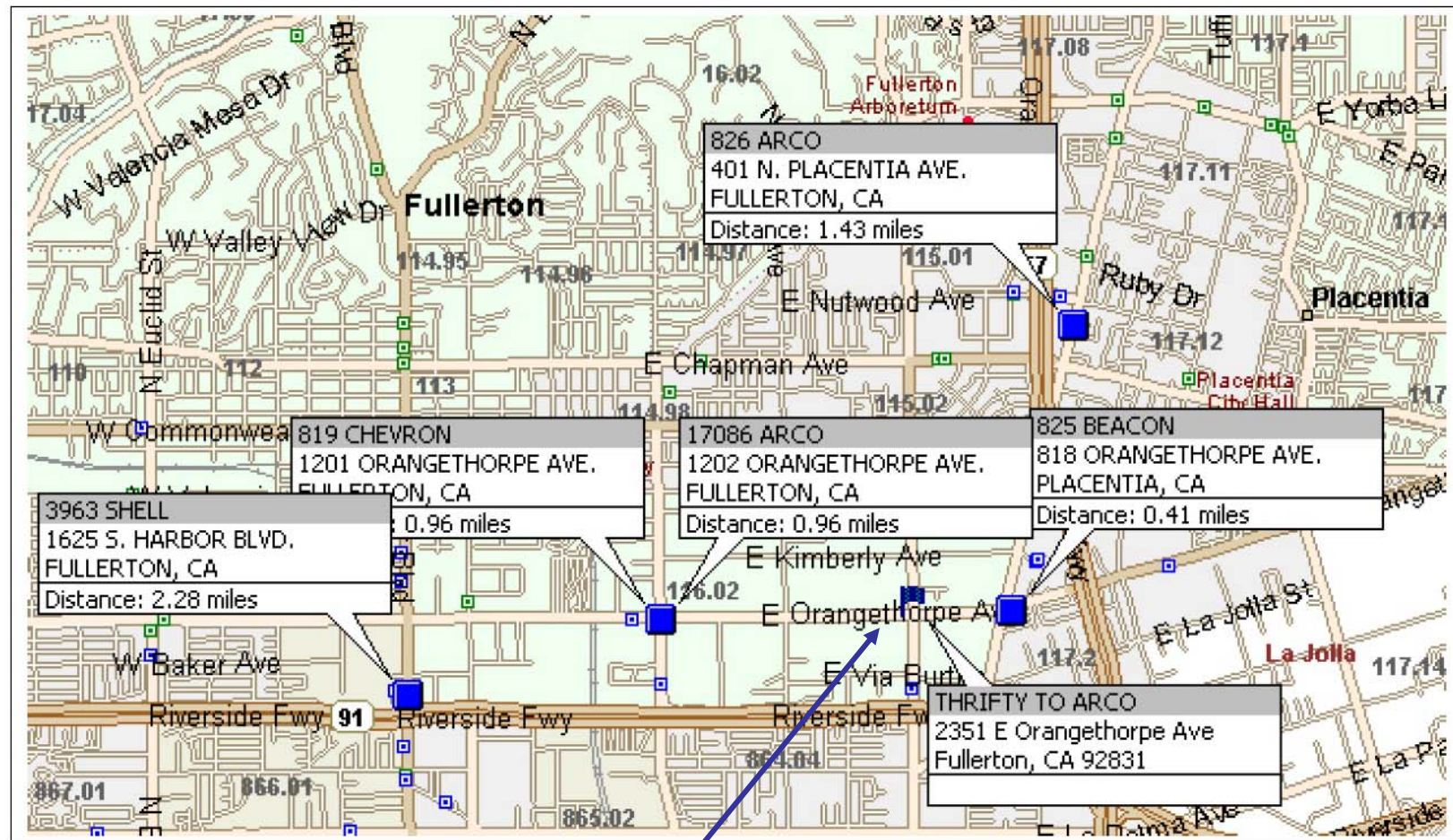
# Empirical Strategy

- Measure the “treatment effect” of a station being “treated” with a competitor who is indep, co-op, or dealer owned
- Possible selection/endogeneity concerns
  - ARCO didn’t randomly assign 260 stations as either co-op or dealer-owned
    - E.g. co-op in less elastic markets
  - Alternative: use discrete change (with panel data) to hopefully control for unobservables
    - Scenario that would create bias: ARCO assigned co-op in markets with low price elasticity, and decide to practice more price discrimination

# Data

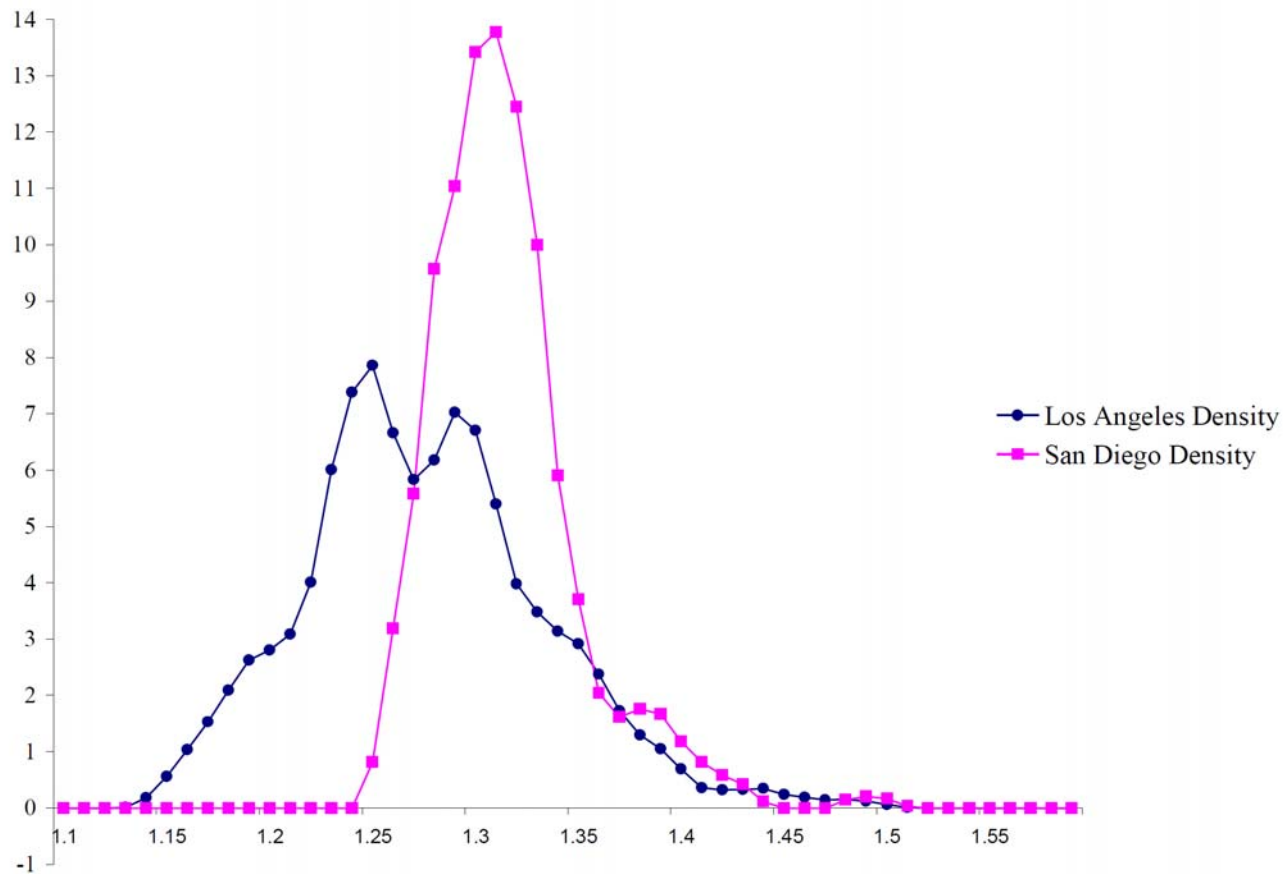
- Census of retail stations in San Diego & LA
- For 20% of stations, sales volumes for Feb, June, Oct, Dec 1997 & 1 price obs/month
  - Very unique data!
- Market definition
  - A station competes with a Thrifty if there's a Thrifty within 1 mile along street

Figure 4: Sample Thrifty Conversion in Fullerton. The flag denotes a Thrifty station that was converted to an ARCO station. The boxes mark the locations of stations with price observations.



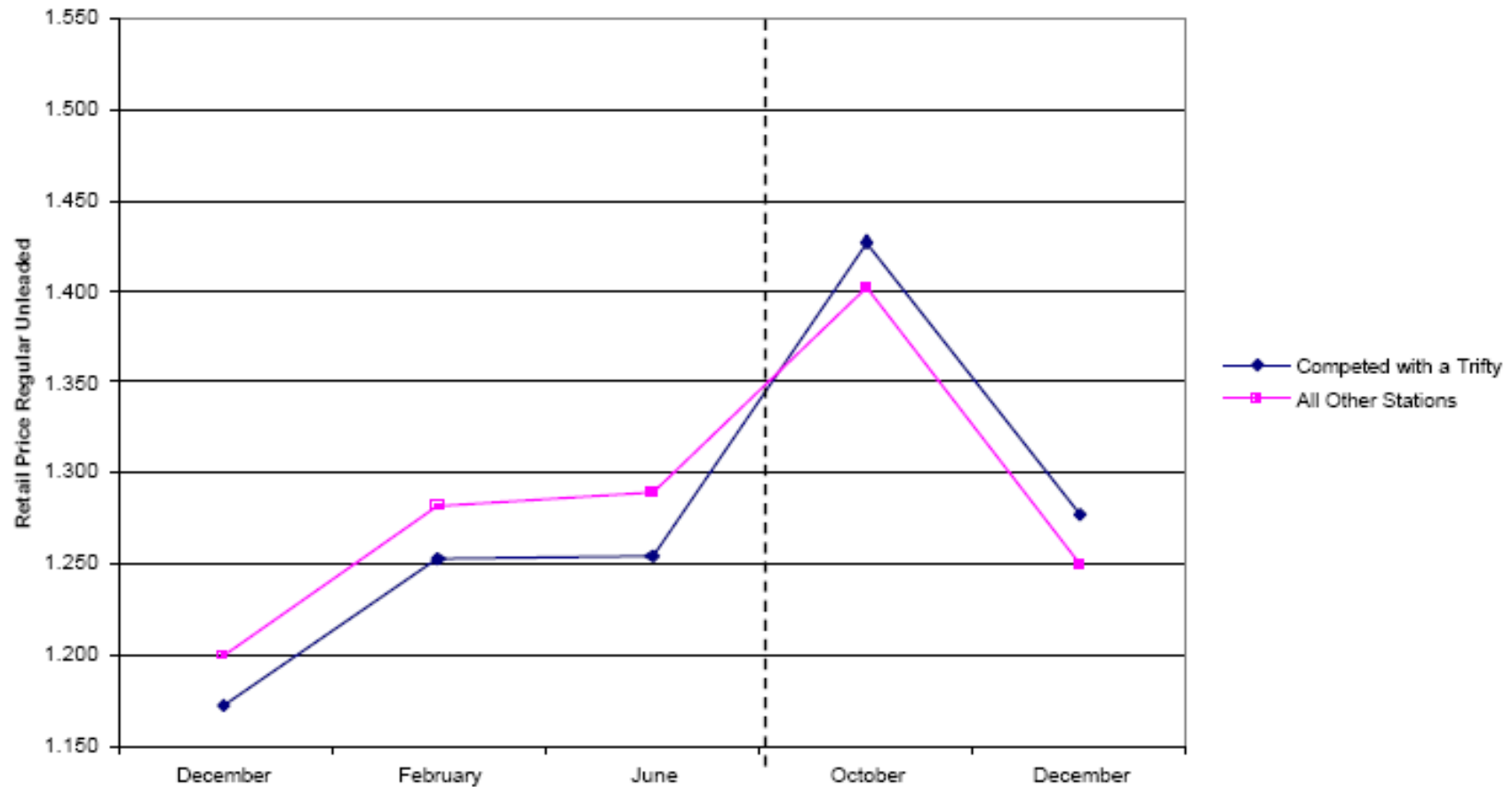
# Heterogeneity Across Stations

Figure 2: Kernel Density Estimates for Retail Price of Regular Unleaded Gasoline in February 1997 Observations for Los Angeles and San Diego

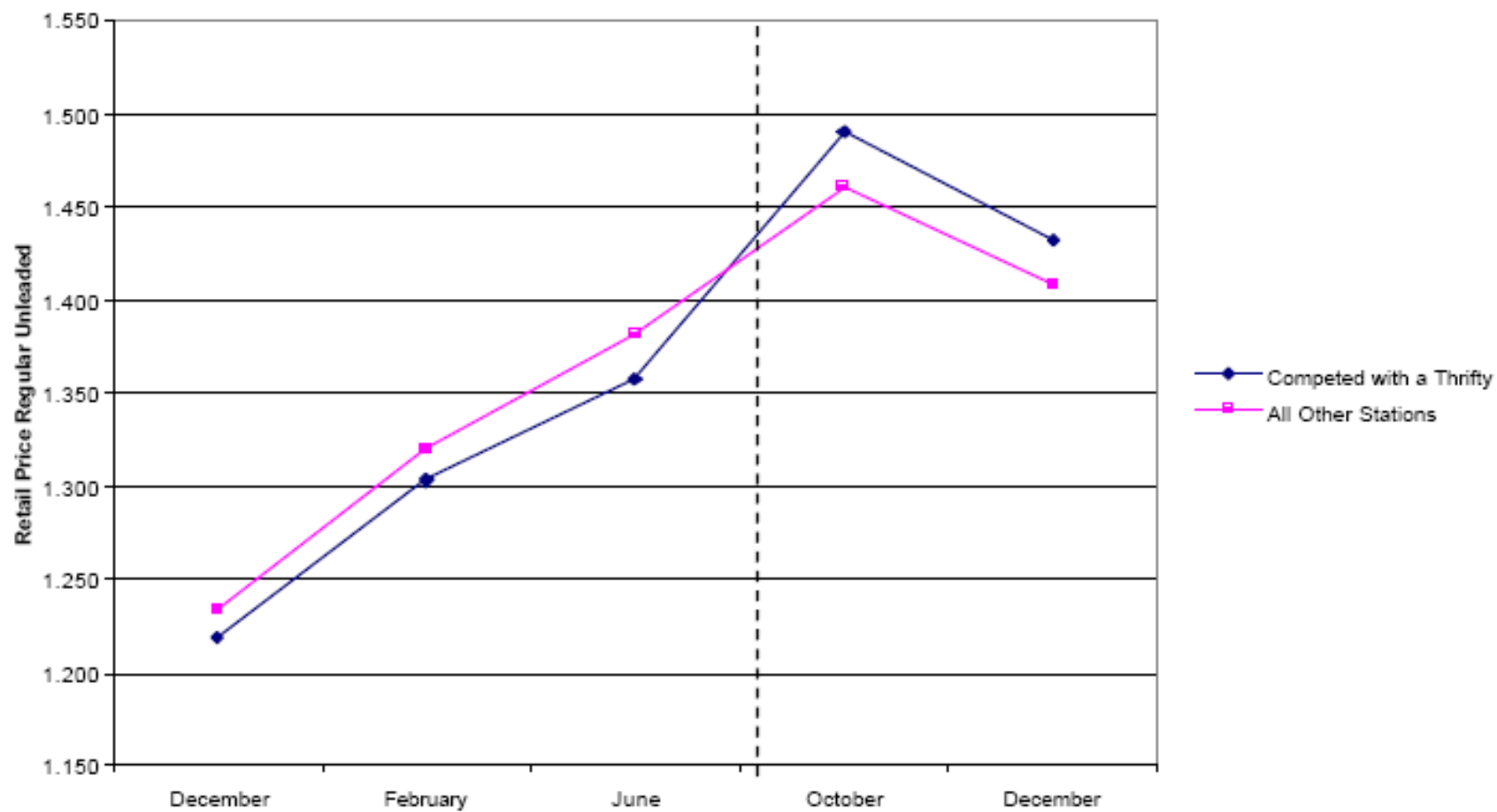


\* Epanechnikov kernel function was used with bandwidth = 0.0095.

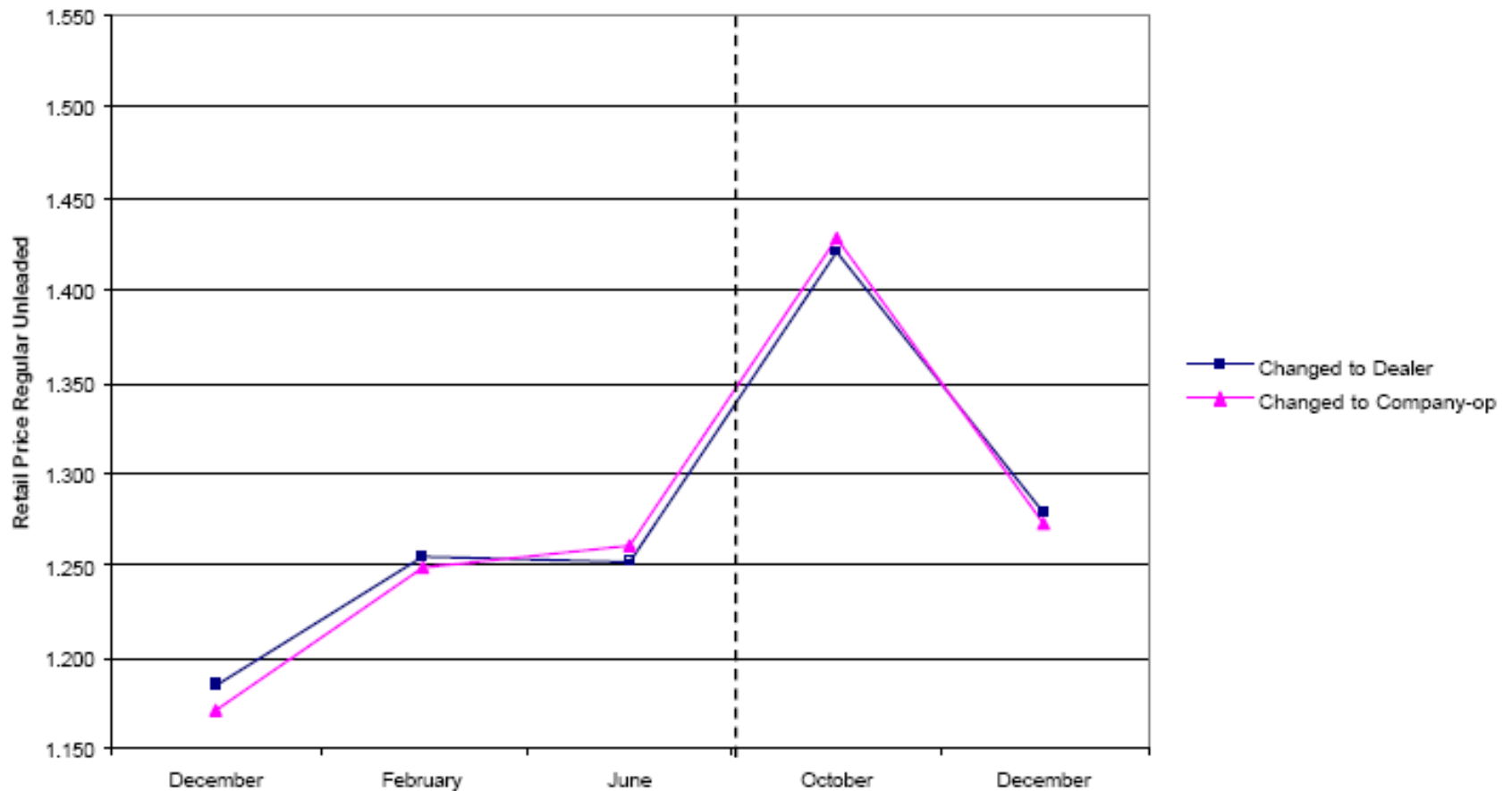
Graph I.a: Los Angeles Treatment and Control Graph



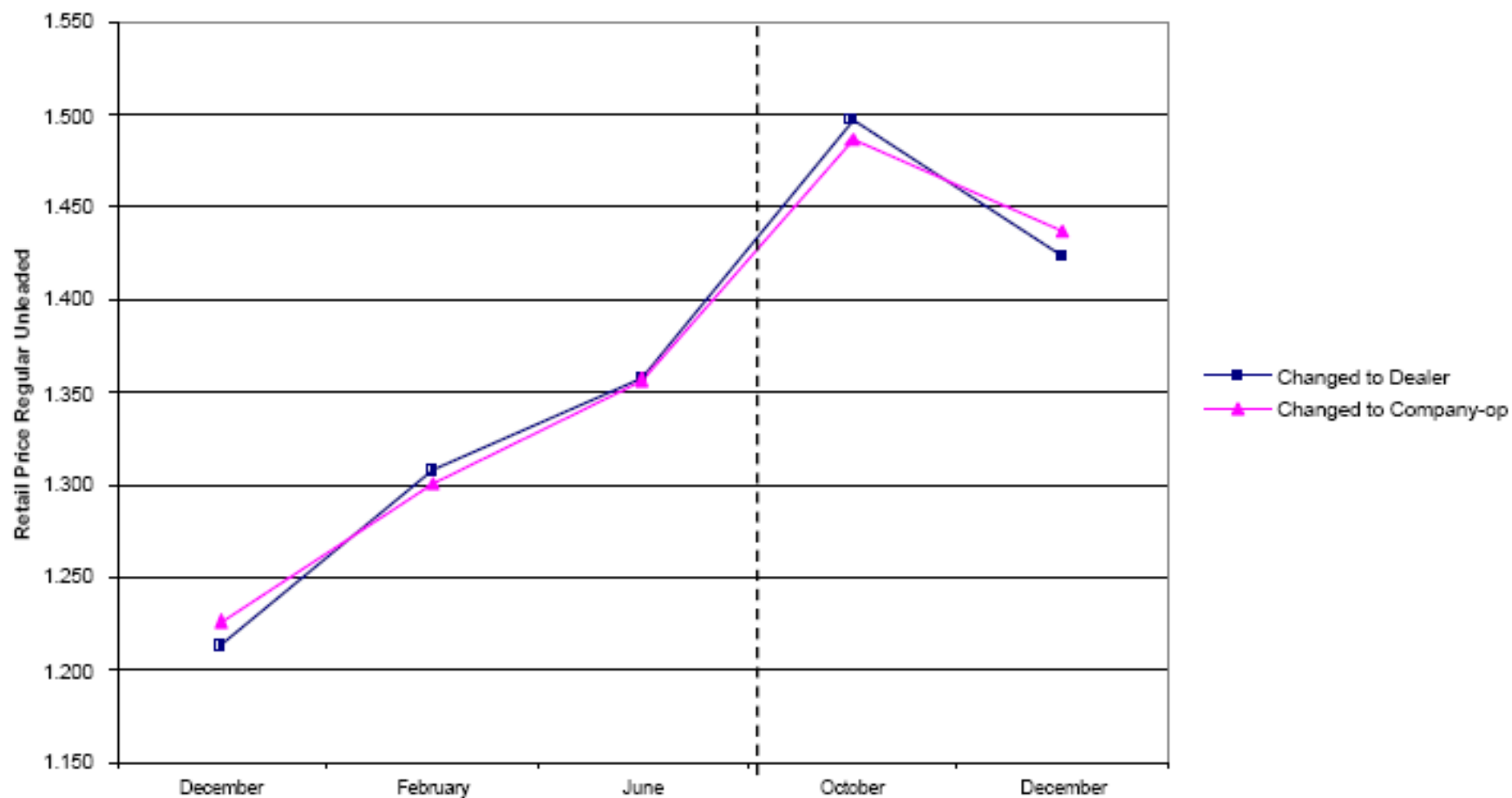
Graph I.b: San Diego Treatment and Control Graph



Graph II.a: Los Angeles Change to Company-op vs. Change to Dealer-run



Graph II.b: San Diego Change to Company-op vs. Change to Dealer-run



# Key Identifying Assumption

- For a standard Diff-in-Diff?
  - The trend for control group is identical for trend in treated absent treatment.
  - How do we see this?
- Other standard diff-in-diff specification tests?
  - Estimate leading dummy indicator as a placebo test

# Results, More Formally

$$p_{it} = \mu + \alpha_i + \delta\gamma + \phi c_{it} + \theta z_{it} + \varepsilon_{it}$$

where  $\alpha_i$  = station effect,  $\gamma$  = city dummy,

$z_{it} = 1(\text{compete w/ indep})$

$c_{it} = 1(\text{compete w/ co - op})$

Indep → 5¢ decrease in market price  
Co-op does not affect market price

Table II: Fixed-Effects Estimation

Dependent Variable: Retail Price for Regular Unleaded

| Variable                         | (1)                | (2)                 | (3)                 |
|----------------------------------|--------------------|---------------------|---------------------|
| Intercept                        | 1.3465<br>(0.0421) | 1.3465<br>(0.0415)  | 1.3617<br>(0.0287)  |
| Company Operated                 | 0.1080<br>(0.0107) | -0.0033<br>(0.0178) | -0.0033<br>(0.0122) |
| Independent                      | -                  | -0.1013<br>(0.0143) | -0.0500<br>(0.0101) |
| LA*February                      | -                  | -                   | 0.0180<br>(0.0065)  |
| LA*June                          | -                  | -                   | 0.0243<br>(0.0065)  |
| LA*October                       | -                  | -                   | 0.1390<br>(0.0064)  |
| SD*February                      | -                  | -                   | -0.0851<br>(0.0036) |
| SD*June                          | -                  | -                   | -0.0304<br>(0.0036) |
| SD*October                       | -                  | -                   | 0.0545<br>(0.0036)  |
| Adj. R-Square                    | 0.3772             | 0.3953              | 0.7181              |
| F-Test for No Fixed Effects:     |                    |                     |                     |
| Numerator DF: 668                |                    |                     |                     |
| Denominator DF: 1999             |                    |                     |                     |
| F value: 3.262                   |                    |                     |                     |
|                                  |                    |                     | Prob.>F: 0.000      |
| Hausman Test for Random Effects: |                    |                     |                     |
| Hausman's M Value: 622.296       |                    |                     | Prob. >M: 0.000     |

# “Replication” Using Different Data

- See Taylor, Kreisle & Zimmerman (AER, 2010) for a less than successful “replication”
  - Another dataset: Oil Price Information Service (OPIS): price data from everytime a client uses a Wright Express fleet card (more frequent data but possibly non-random)
- Find that the Hastings 5¢ effect is an order of magnitude smaller using OPIS data

# Key Methodological Characteristics

- Assumptions Necessary
  - Quasi-random assignment (or at least assignment not correlated w/ pricing policy)
  - Treatment effect is same across station (“average treatment effect”)
- Assumptions Not Necessary
  - Explicit model of demand function for differentiated product
  - A strategic model of competition

# Aside: Other Recent Research on Gasoline Markets

- Pricing: Matt Lewis (Ohio State), Michael Noel (UCSD)
- Effect of regional regulation: Erich Muehlegger (KSG)
- Gasoline Prices & car purchase behavior: Hunt Allcott (MIT), Mark Jacobsen (UCSC), Meghan Busse (Northwestern)

**TABLE 5 Predicted Percent Change in Prices and Quantities as a Result of Mergers**

|                       | Post and Nabisco |          | GM and Nabisco |          | GM and Chex |          | Kellogg and Quaker Oats |          | GM and Quaker Oats |          |
|-----------------------|------------------|----------|----------------|----------|-------------|----------|-------------------------|----------|--------------------|----------|
|                       | <i>p</i>         | <i>q</i> | <i>p</i>       | <i>q</i> | <i>p</i>    | <i>q</i> | <i>p</i>                | <i>q</i> | <i>p</i>           | <i>q</i> |
| K Corn Flakes         | .0               | .0       | .0             | .1       | .0          | .1       | .0                      | .5       | .0                 | .3       |
| K Raisin Bran         | .1               | .1       | .1             | .3       | .1          | .2       | 1.4                     | -1.7     | .5                 | .7       |
| K Frosted Flakes      | .0               | .0       | .0             | .1       | .0          | .1       | .3                      | -.4      | .1                 | .3       |
| K Rice Krispies       | .0               | .1       | .1             | .2       | .1          | .4       | 5.1                     | -4.1     | .7                 | 2.0      |
| K Frosted Mini Wheats | .0               | .2       | .0             | .2       | .1          | .3       | 2.7                     | -4.1     | .3                 | 2.9      |
| K Froot Loops         | .0               | .1       | 0              | .2       | .1          | .5       | 9.3                     | -15.3    | .7                 | 8.0      |
| K Special K           | .0               | .1       | .1             | .2       | .0          | .2       | .2                      | .2       | .1                 | .4       |
| K NutriGrain          | .0               | .0       | .1             | .1       | .0          | .1       | .0                      | .4       | .1                 | .3       |
| K Crispix             | .0               | .1       | .0             | .2       | .1          | .4       | 3.4                     | -3.8     | .5                 | 2.7      |
| K Cracklin Oat Bran   | .0               | .1       | .0             | .2       | .0          | .4       | 3.4                     | -6.8     | .4                 | 3.7      |
| GM Cheerios           | .0               | .2       | .7             | -.9      | 1.1         | -1.3     | .5                      | 1.3      | 4.1                | -3.5     |
| GM Honey Nut Cheerios | .0               | .1       | .5             | -.6      | .8          | -.9      | 1.0                     | 3.2      | 11.5               | -11.2    |
| GM Wheaties           | .0               | .0       | .0             | 0        | .1          | -.1      | .1                      | .5       | .1                 | .3       |
| GM Total              | .0               | .1       | .3             | -.8      | .2          | -.6      | .1                      | .4       | .2                 | .1       |
| GM Lucky Charms       | .0               | .1       | .3             | -.4      | .7          | -.8      | .8                      | 3.3      | 9.3                | -10.6    |
| GM Trix               | .0               | .1       | .3             | -.3      | .7          | -.9      | .7                      | 3.5      | 8.6                | -9.6     |
| GM Raisin Nut         | .0               | .2       | .4             | -.7      | .5          | -.9      | .3                      | 1.5      | 1.8                | -2.7     |
| P Raisin Bran         | .9               | -1.5     | .0             | .5       | .0          | .4       | .1                      | 1.5      | .2                 | 1.7      |
| P Grape Nuts          | 1.5              | -2.8     | .1             | .7       | .0          | .4       | .1                      | 2.3      | .1                 | 3.0      |
| Q 100% Natural        | .0               | .1       | .0             | .3       | .0          | .5       | 10.2                    | -17.0    | 11.4               | -19.3    |
| Q Life                | .0               | .1       | .0             | .3       | .1          | .5       | 15.5                    | -16.7    | 23.8               | -25.3    |
| Q CapNCrunch          | .0               | .1       | .0             | .3       | .1          | .4       | 16.8                    | -16.7    | 29.1               | -30.9    |
| R Chex                | .0               | .2       | .0             | .3       | 12.2        | -19.0    | .0                      | 2.1      | .1                 | 3.4      |
| N Shredded Wheat      | 3.1              | -8.6     | 7.5            | -18.8    | .0          | .4       | .0                      | 1.9      | .0                 | 2.5      |

Figures are the median change for each brand over the 45 cities in the last quarter of 1992, and are based on Table 2.

# Estimated Changes in Consumer Surplus

**TABLE 7**      **Change in Variable Profits and Consumer Surplus as a Result of Mergers (millions of dollars per year)**

|                         | Post and Nabisco | General Mills and Nabisco |
|-------------------------|------------------|---------------------------|
| <b>Consumer surplus</b> | −13.98           | −26.79                    |

Where did these numbers from come?

We'll talk later in the course about a result from Small and Rosen (EMA, 1981) that provides a closed-form formula for consumer surplus under a certain distributional assumption (logit)

➔ A very powerful result if the assumption is reasonable

# Differences in Two Approaches to Estimating Effects of Mergers

- Data “the world gave us”
- Empirical Assumptions
- “Bottom line” estimates



# Is the Effect of Independents Due to a Product Differentiation?

Table III: Fixed-Effects Estimation, Independent coefficient by Brand Group  
Dependent Variable: Retail Price for Regular Unleaded (Standard Errors in Parentheses)

| Variable                          | (1)<br>Parameter Estimate | (2)<br>Parameter Estimate |
|-----------------------------------|---------------------------|---------------------------|
| Intercept                         | 1.3622<br>(0.0287)        | 1.3620<br>(0.0287)        |
| Company Operated                  | -0.0018<br>(0.0124)       | -0.0008<br>(0.0124)       |
| Independent · High-share Brands   | -0.0273<br>(0.0125)       | -0.0362<br>(0.0156)       |
| Independent · Middle-share Brands | -0.0530<br>(0.0154)       | -0.0617<br>(0.0179)       |
| Independent · Low-share Brands    | -0.0700<br>(0.0185)       | -0.0741<br>(0.0190)       |
| Independent · ARCO                | -0.0731<br>(0.0149)       | -0.0741<br>(0.0149)       |
| Independent · N Decreased         | ---                       | 0.0130<br>(0.0136)        |
| City-time Effects                 | Yes                       | Yes                       |
| Adj. R-Square                     | 0.7183                    | 0.7187                    |

High=Chevron,Shell,Unocal Mid=Exxon,Mobil,Texaco Low=Beacon, Circle K...

➔ Independents compete more with “low brand loyalty” brands